

Business Management System (BMS)

Product Blueprint v1.0

Version: 1.0

Status: Master Product Specification

0. Executive Summary

Product Overview

The Business Management System (BMS) is a configurable, multi-tenant business platform that enables businesses of different sizes and industries to manage their operations through a single, unified system. Unlike traditional business software that exposes every feature to every customer, BMS configures itself around each business through workspaces, capabilities, and role-aware experiences. The software adapts to the business, not the other way around.

Core Philosophy

- Software adapts to the business.
- Businesses grow by enabling capabilities rather than replacing software.
- Every interface supports meaningful business work.
- Users only see what is relevant to them.
- Consistency creates confidence.

Primary Users: Owner, Manager, Cashier. **Future:** Accountant, Auditor, Supplier Portal User, Customer Portal User.

Core Architecture

Workspace → Capabilities → Permissions → Navigation → Pages Components → Actions

Everything inside BMS is derived from this pipeline (fully specified in Section 14, Feature Availability Model).

Primary Business Domains: Selling, Inventory, Purchasing, Suppliers, Customers, Employees, Reporting, Business Configuration.

Design Philosophy: The interface should always be calm, professional, predictable, business-first, enterprise-quality, and progressive — never louder or more complex than the task in front of the user requires.

PART I — Product Foundation

1. Product Vision

The Business Management System (BMS) is a modular, configurable platform designed to simplify the daily operations of small and medium-sized businesses while remaining capable of supporting larger organizations with multiple businesses and branches. Unlike traditional business software that forces every organization into the same workflow, BMS is designed to adapt to the unique structure and operational needs of each business. From a small retail shop managing a single register to an enterprise operating multiple businesses across several locations, the platform scales through configurable capabilities rather than entirely different products.

The primary goal of BMS is to help businesses spend less time managing software and more time running their operations.

Every design decision within the platform is guided by four objectives:

- **Operational Efficiency** — Every screen should reduce the effort required to complete a business task. The system exists to support work rather than showcase information.
- **Business-First Thinking** — Interfaces should reflect how business owners think about their operations, not how software developers organize databases or system architecture.
- **Progressive Growth** — Businesses should be able to begin with only the capabilities they need and expand their workspace as their operations become more sophisticated without migrating to another system.
- **Confidence Through Clarity** — The system should always communicate what is happening, why it matters, and what the user can do next. Clear feedback, predictable interactions, and consistent navigation build confidence and reduce errors.

BMS is therefore not simply a collection of business modules. It is an operational workspace that grows with the business while maintaining simplicity, consistency, and efficiency.

Workspace & Capability Philosophy

One of the defining principles of BMS is that the software should adapt to the business rather than requiring the business to adapt to the software.

Businesses differ significantly in their operations. A supermarket, pharmacy, restaurant, wholesaler, manufacturer, school, or service company may all require different capabilities despite sharing common administrative functions.

Rather than exposing every feature to every business, BMS organizes functionality into configurable workspaces and capabilities.

Workspace — A curated collection of capabilities designed around a particular style of business operation. Examples: Essentials, Retail Pro, Multi-Branch, Enterprise Operations, Custom Workspace. Each workspace provides an appropriate starting point while allowing businesses to evolve over time. The selected workspace determines the initial capabilities available to the business but does not permanently restrict future expansion.

Capabilities — Functional areas that enable specific business operations: Sales & Checkout, Products & Inventory, Purchasing, Supplier Management, Customer Management, Employee & Role Management, Business Analytics, Multi-Branch Management, Business Configuration. Capabilities represent what the business can do rather than how the software is organized internally.

Configuration Rather Than Custom Development — Businesses are expected to differ. Instead of creating separate editions of the software, BMS allows organizations to enable or disable capabilities based on their operational needs. This results in simpler interfaces, lower cognitive load, faster onboarding, improved usability, and better scalability.

Progressive Expansion — Businesses should never feel they selected the wrong workspace. As business requirements change, additional capabilities can be enabled from Business Settings without data migration or system replacement.

Role-Specific Experiences — Capabilities determine which features exist within a business. Permissions determine who may access those capabilities. Owners see strategic information and configuration options. Managers see operational tools and daily oversight. Cashiers see only the selling workspace required for customer transactions. Navigation, dashboards, and workflows are therefore determined by both the enabled capabilities and the user's assigned role.

Design Implications — Every interface within BMS should expose only enabled capabilities, avoid displaying inaccessible functionality, speak the language of business operations, support progressive discovery, and minimise unnecessary complexity. The result is a system that feels intentionally designed for each business rather than generically built for every possible business.

2. Product Goals

- **Simplicity** — A business only ever sees the capabilities it has enabled (Section 11); nothing is exposed "in case it's needed later."

- **Scalability** — Growth happens by enabling capabilities and branches, never by migrating to a different product or re-onboarding (Section 10.7).
- **Consistency** — The same module anatomy (Section 9.7), the same interaction language, the same status/color conventions apply everywhere, so learning one part of BMS teaches the rest.
- **Efficiency** — Every workflow is measured against clicks, keystrokes, and cognitive load, especially in high-frequency workspaces (POS, Inventory) where seconds compound across hundreds of repetitions a day.

3. Design Philosophy

- **Software Adapts to the Business** — The business's structure and needs shape the software's configuration; never the reverse.
- **Progressive Complexity** — A single-shop Essentials business should never feel like it's staring at enterprise software, and an Enterprise Operations business should never feel capped by a "small business tool." Same platform, different resolved experience (Section 9.1.1).
- **Role Awareness** — Every interface is a function of workspace, enabled capabilities, role, and permissions simultaneously — not a single static layout with things hidden.
- **Recognition Over Recall** — Autocomplete, visible history, and Command Search over memorized codes or paths.

4. Product Principles

Business First · Consistency · Progressive Disclosure · Recognition over Recall · One Primary Action per Screen · Context Preservation · Actionable Interfaces (a screen without a next action is missing a task, not just missing data) · Operational Efficiency

5. Target Users

User	Status	Notes
Owner	Current	Full authority, self-registers, whole-business view
Manager	Current	Branch-scoped operational authority
Cashier	Current	Sales & Checkout only, highest-frequency user
Accountant	Future	Likely a custom role scoped to Reports + Taxes & Finance, read-heavy
Auditor	Future	Read-only, cross-branch, audit-log-focused
Supplier Portal User	Future	External identity, scoped to their own POs and performance data only
Customer Portal User	Future	External identity, scoped to their own purchase history/loyalty

6. Business Types

Type	Operational difference from baseline retail
Retail	Baseline model
Wholesale	Bulk units, tiered/negotiated pricing per customer, larger average order size
Pharmacy	Regulatory/batch and expiry tracking, controlled-item sale restrictions
Hardware	Wide SKU variance, frequent custom/cut-to-order items, supplier-sourced special orders
Restaurant	Order-then-prepare flow rather than scan-and-pay; table/seat concept; different POS shape entirely

Type	Operational difference from baseline retail
Service Business	Sells time/labor, not stock — Products & Inventory may be entirely disabled
Manufacturing (future)	Bill-of-Materials concept and raw-material-to-finished-good inventory transformation

7. Business Structures

Organization → Business (workspace assigned here — Section 10.2) → Branch

- **Single Shop** — One Business, one Branch. Multi-Branch Management capability stays hidden entirely.
- **Retail Chain** — One Business, many Branches, sharing one workspace and capability set.
- **Holding Company** — One Organization, several Businesses, each independently workspace-configured.

PART II — Product Architecture

8. Information Architecture

1. Public Pages (Before Authentication)

```

Landing Page
├── Features
├── Pricing
├── Business Types
├── Business Capabilities
├── About
├── Contact
├── Documentation
├── Sign In

Authentication
├── Login
├── Forgot Password
├── Reset Password
├── Accept Employee Invitation
├── Create Business Account

```

2. Business Onboarding

```

Business Onboarding
├── Business Details
├── Choose Workspace
├── Compare Workspaces
├── Explore All Capabilities
├── Customize Workspace (optional)
├── Business Provisioning
├── Onboarding Complete

```

3. Owner Workspace

```

Owner Dashboard
├── Business Performance
├── Needs Attention
├── Recent Activity

```

- └ Business Insights
- └ Quick Actions
- └ Branch Selector

4. Cashier Workspace (POS)

- POS Workspace
 - └ Ready to Sell
 - └ Live Search
 - └ Shopping Cart
 - └ Customer Selection
 - └ Payment
 - └ Receipt
 - └ Returns
 - └ Resume Held Sale
 - └ Offline Mode
 - └ Transaction Complete
 - └ Held Sales
 - └ Empty
 - └ Single Held Sale
 - └ Multiple Held Sales
 - └ Offline States
 - └ Offline Banner
 - └ Queue Pending
 - └ Sync Complete
 - └ Sync Failed

5. Manager Workspace

- Manager Dashboard
 - └ Store Summary
 - └ Needs Attention
 - └ Inventory Status
 - └ Deliveries
 - └ Returns
 - └ Operational Insights
 - └ Recent Activity
 - └ Quick Actions

6. Product Management

- Products
 - └ Product List
 - └ Product Details
 - └ Add Product
 - └ Edit Product
 - └ Categories
 - └ Brands
 - └ Product Variants
 - └ Barcode Management
 - └ Product Archive

7. Inventory Management

- Inventory
 - └ Inventory Overview
 - └ Stock Levels
 - └ Stock Adjustments
 - └ Stock Count

- └─ Inventory History
- └─ Low Stock
- └─ Out of Stock
- └─ Inventory Valuation
- └─ Stock Transfers

8. Supplier Management

Suppliers

- └─ Supplier List
- └─ Supplier Details
- └─ Add Supplier
- └─ Edit Supplier
- └─ Supplier Products
- └─ Purchase History
- └─ Supplier Performance

9. Purchase Orders

Purchase Orders

- └─ Purchase Order List
- └─ Draft PO
- └─ Create Purchase Order
- └─ Purchase Order Details
- └─ Approvals
- └─ Goods Receiving
 - └─ Partially Received
 - └─ Completed
 - └─ Cancelled

10. Customer Management

Customers

- └─ Customer List
- └─ Customer Details
- └─ Add Customer
- └─ Edit Customer
- └─ Purchase History
- └─ Loyalty
- └─ Store Credit (Future)

11. Sales & Returns

Sales

- └─ Sales History
- └─ Sale Details
- └─ Returns
 - └─ Return Details
- └─ Void Sale
- └─ Refund History

12. Reports & Analytics

Reports

- └─ Dashboard
- └─ Sales Reports
- └─ Inventory Reports
- └─ Customer Reports

- Supplier Reports
- Purchase Reports
- Employee Reports
- Branch Reports
- Profit Analysis
- Export Reports
- Scheduled Reports

13. User Management

- Employees
 - Employee List
 - Employee Details
 - Invite Employee
 - Roles
 - Permissions
 - Activity
 - Access History

14. Business Settings

- Business Settings
 - Business Profile
 - Workspace
 - Branches
 - Taxes & Finance
 - Receipts
 - Hardware
 - Integrations
 - Notifications
 - Security
 - Subscription
 - Audit Log

15. Notification Center

- Notification Center
 - All
 - Needs Attention
 - Inventory
 - Purchasing
 - Sales
 - Employees
 - System
 - Archived
 - Notification Preferences

16. Help & Support

- Help & Support
 - Getting Started
 - Business Capabilities
 - Tutorials
 - User Guides
 - Troubleshooting
 - Contact Support
 - System Status

- └─ Release Notes
- └─ Search

17. User Profile

- User Profile
 - └─ Personal Information
 - └─ Security
 - └─ Password
 - └─ Two-Factor Authentication
 - └─ Notification Preferences
 - └─ Active Sessions
 - └─ Appearance

18. Supporting Screens

- System Pages
 - └─ Empty States
 - └─ Loading States
 - └─ Confirmation Dialogs
 - └─ Success Messages
 - └─ 403 Forbidden
 - └─ 404 Not Found
 - └─ 500 Internal Server Error
 - └─ Offline Screens

Overall Site Map

- Landing Page
- Authentication (Login, Forgot Password, Create Business)
- Onboarding (Business Details, Workspace Selection, Compare Workspaces, Explore Capabilities, Provisioning)
- Application
 - └─ Owner Dashboard
 - └─ Manager Dashboard
 - └─ POS Workspace
 - └─ Products
 - └─ Inventory
 - └─ Suppliers
 - └─ Purchase Orders
 - └─ Customers
 - └─ Sales & Returns
 - └─ Reports
 - └─ Employees
 - └─ Business Settings
 - └─ Notifications
 - └─ Help & Support
 - └─ User Profile
 - └─ System Pages

8A. Product Scope

In Scope (MVP): Authentication, Business Onboarding, Workspace Configuration, Point of Sale, Products, Inventory, Suppliers, Purchase Orders, Customers, Sales & Returns, Reports, Employees, Business Settings, Notifications, Help, User Profiles.

Planned: CRM, Promotions, Loyalty, Expenses, Accounting, AI Insights, Forecasting.

Out of Scope (Current): Payroll, HR Management, Fleet Management, Manufacturing, ERP, Accounting Ledger, E-Commerce Platform. These remain future expansion areas explicitly not implied by anything in the current architecture, even where the capability model would technically support adding them later.

8B. Product Success Metrics

- **User Experience:** Average onboarding time, time to first completed sale, time to add a new product, time to create a purchase order.
- **Operational:** Average sale duration, inventory adjustment accuracy, purchase approval turnaround, stock count completion rate.
- **Product:** Daily active users, monthly active businesses, workspace adoption, capability adoption, feature discoverability.
- **Business:** Reduced training time, reduced stock discrepancies, faster checkout, better purchasing decisions.

9. Navigation Architecture

9.1 Governing principles

Navigation in BMS is shaped by two things simultaneously: which capabilities are enabled for the business, and which role the current user holds. A nav item only appears if both conditions are satisfied — enabled and permitted.

9.1.1 Capability-Driven Navigation Resolution

Navigation isn't hand-authored per business — it's resolved through a fixed pipeline:

Workspace → Enabled Capabilities → Role Permissions → Feature Flags (future) → Navigation Tree

- **Workspace** sets the starting bundle of capabilities at provisioning.
- **Enabled Capabilities** — Business-controlled, toggled in Business Settings. Determines what exists for this business.
- **Role Permissions** — Determines who on this business's team can see and act on what exists.
- **Feature Flags** — Platform-controlled, for staged rollout or beta access. Distinct from capabilities: a disabled capability means "this business chose not to use this"; a disabled feature flag means "this hasn't been released to this business yet."
- **Navigation Tree** — The final, resolved output rendered to the user.

This is what future-proofs BMS: adding "Warehouse Management," "Manufacturing," "Repairs," or "Asset Tracking" as new capabilities under Inventory requires zero navigation architecture changes — only new entries in the capability layer.

9.2 Navigation shells (one per role)

- **Owner Shell** — Full sidebar, all enabled-capability modules, branch selector, strategic framing.
- **Manager Shell** — Narrower sidebar scoped to daily operations, expandable via permissions.
- **Cashier Shell (POS)** — No persistent sidebar. Full-screen workspace, minimal chrome.

9.3 Global Top Bar & Navigation Context

Persistent across every page in the Owner and Manager shells:

- **Navigation Context strip** — A constant, always-visible answer to Where am I? Which business? Which workspace? Which branch?, e.g.: *ABC Supermarket · Retail Pro · Westlands Branch → Inventory → Products → Cooking Oil*
- **Command Search** (9.10)
- **Notification bell** opens the Notification Center
- **Help (?)** — Contextual to current module

- **Branch selector** — Part of the context strip
- **Profile avatar** opens User Profile

The POS shell omits this entirely except a minimal status strip (till/terminal identity, offline indicator, cashier name).

9.4 Primary (sidebar) navigation — Owner Shell

Grouped into labeled sections rather than ten equal-weight items (Hick's Law):

Overview

- Dashboard

Operations

- Products
- Inventory
- Purchase Orders
- Suppliers
- Customers
- Sales & Returns

Insights

- Reports

Administration

- Employees
- Business Settings

9.5 Primary navigation — Manager Shell

Overview

- Dashboard (Manager view)

Operations

- Inventory
- Deliveries
- Returns
- Products (if permitted)

Manager access to Reports, Employees, and Business Settings is a permission question, not a navigation question.

9.6 Secondary Navigation, Breadcrumbs & Location Trail

- **Tabs** for sibling views of the same entity (e.g. Employees: List / Roles / Permissions / Activity).
- **Breadcrumbs** for drill-down hierarchy (e.g. Suppliers → Supplier Details → Purchase History).

These extend the Navigation Context strip (9.3) as one continuous line, not a separate mechanism. Purchase Orders' lifecycle states are filter tabs on one list view, not separate pages.

9.7 Module Anatomy

Every list-based module shares the same internal structure:

Header → Toolbar → Filters → Table → Pagination

A strict consistency rule: once a user learns Products, they already know how to operate Suppliers.

9.8 Cross-cutting entry points & Notifications

Notification Center, Help & Support, User Profile — reachable from anywhere, not tied to any one capability.

Notifications split into two types:

- **Passive** — Informational (sale completed, inventory synced). Expire naturally.
- **Actionable** — Require a decision (approve PO, low stock, employee invitation, subscription expiring). Persist until resolved; visually distinct from passive (Von Restorff effect).

9.9 Workspace & Branch Switching

The branch selector re-scopes the current page's data in place. "All Branches" is valid wherever aggregate reporting makes sense, not where it isn't (POS, Stock Adjustments).

Rule: Switching branch never silently discards work. Unsaved changes trigger Save / Discard / Stay Here.

Workspace type is configured once in Business Settings and determines which capabilities exist — provisioning, not navigation.

9.10 Global Search → Command Search

One input, many result types (entities: Product, Customer, Supplier, PO, Employee, Branch, Setting, Help Article; actions: "Create Product," "Go to Reports") in the spirit of Linear, Notion, Slack. This is the same surface as the Command Palette overlay — one unified system.

9.11 Navigation Persistence (post-MVP)

BMS should remember, per user: last selected branch, expanded/collapsed sidebar groups, last-used report filters, preferred table density, preferred sort order, collapsed panel states.

9.12 Mobile Adaptation

Bottom tab bar, capability-aware — prioritize, never simply hide:

- **Owner:** Home · Products · Inventory · Sales · More
- **Retail Pro:** Home · POS · Products · Inventory · More
- **Enterprise:** Home · Reports · Branches · Employees · More

9.13 Depth Guideline

No page more than three steps from its shell's home: Sidebar item → List → Detail.

9.14 Navigation Behaviour Rules

- **Preserve Context** — Retain branch, filters, date ranges across module changes.
- **Never Lose Work** — Warn before leaving pages with unsaved changes.
- **Stable URLs** — Every page is bookmarkable.
- **Progressive Disclosure** — Advanced actions appear only when relevant.
- **Keyboard Navigation** — Supported for high-frequency tasks, especially POS and inventory.
- **Predictable Navigation** — The same action always leads to the same destination.

10. Workspace Architecture

10.1 What a Workspace Is

A named, curated bundle of capabilities giving a business a sensible starting configuration — a starting point, not a permanent ceiling.

10.2 Workspace Assignment Scope

A workspace is assigned per Business, not per Organization and not per Branch. An Organization can contain multiple Businesses, each selecting its own workspace independently. All Branches under a Business share that Business's workspace and capability set.

10.3 Workspace Catalog

Workspace	Intended for	Default capability bundle
Essentials	Single-location shop, simplest operation	Sales & Checkout, Products & Inventory, Business Configuration
Retail Pro	Established single-branch retail business	Essentials + Customer Management, Supplier Management, Purchasing, Business Analytics
Multi-Branch	Business operating 2+ locations	Retail Pro + Multi-Branch Management, cross-branch Reporting
Enterprise Operations	Larger, multi-business or high-complexity operations	Multi-Branch + advanced Employee & Role Management, Audit Log, full Business Analytics
Custom Workspace	Any business whose needs don't map to a preset	No default bundle — every capability chosen individually

10.4 Capability Catalog & Dependencies

Capability	Depends on	Notes
Sales & Checkout	-	Foundational; every workspace includes it
Products & Inventory	-	Foundational
Purchasing	Products & Inventory, Supplier Management	Can't order stock from suppliers that don't exist as a concept
Supplier Management	-	Can exist without Purchasing
Customer Management	-	Independent
Employee & Role Management	-	Present at a basic level in every workspace
Business Analytics	Sales & Checkout	Needs transaction data to report on
Multi-Branch Management	-	Only meaningful with 2+ branches
Business Configuration	-	Always present

10.5 Capability Lifecycle States

- **Enabled** — Active, contributing to the navigation tree.
- **Available** — Not enabled, but the business's plan permits it.
- **Locked** — Requires a plan upgrade.
- **Coming Soon** — Gated by a feature flag, not a plan.

10.6 Onboarding Flow Workspace Selection

Business Details → Choose Workspace → Compare Workspaces → Explore All Capabilities → Customize Workspace (optional) → Business Provisioning → Onboarding Complete

10.7 Changing Workspace After Onboarding

- **Enabling a capability** — Always safe, takes effect immediately.
- **Disabling a capability** — Requires a warning if existing data is tied to it; data preserved, not deleted.
- **Switching workspace type** — Really just enabling several capabilities at once; no re-onboarding, no data migration.
- **Downgrading workspace tier** — Capabilities beyond the new tier move to Locked, not disabled outright.

10.8 Custom Workspace

A Custom Workspace and a heavily-modified preset are indistinguishable at runtime — both are just "a Business with this particular set of enabled capabilities." The workspace name is metadata, not an enforced constraint.

10.9 Relationship to Subscription & Billing

Capability availability = subscription plan $\cap$ workspace choice, resolved before Role Permissions enter the pipeline.

11. Capability Architecture

11.1 Anatomy of a Capability

Every capability is defined by six components:

1. Navigation contribution
2. Data domain
3. Permission set (action-level, not blanket)
4. Dashboard surface
5. Notification types
6. Dependencies

11.2 Capability Catalog (Current)

Capability	Owns (modules)	Key permissions
Sales & Checkout	POS Workspace, Sales & Returns	Process Sale, Void Sale, Issue Refund, Apply Discount
Products & Inventory	Products, Inventory	Create/Edit Product, Adjust Stock, Stock Count, Transfer Stock
Purchasing	Purchase Orders	Create PO, Approve PO, Receive Goods, Cancel PO
Supplier Management	Suppliers	Add/Edit Supplier, View Supplier Performance
Customer Management	Customers	Add/Edit Customer, View Purchase History, Manage Loyalty
Employee & Role Management	Employees	Invite Employee, Assign Role, Edit Permissions, View Activity
Business Analytics	Reports	View Reports, Export Reports, Schedule Reports
Multi-Branch Management	Branch Selector, Settings → Branches	Add Branch, Configure Branch, Cross-Branch Reporting
Business Configuration	Business Settings (core)	Edit Business Profile, Manage Taxes, Manage Receipts, Manage Integrations

11.3 Capabilities vs. Modules

A capability is a business-facing, toggleable concept; a module is the actual screen/IA unit implementing it. Not always one-to-one — disabling Customer Management doesn't break Sales & Checkout, it just removes the customer-selection step.

11.4 Permission Granularity Within a Capability

Permissions are action-level, not capability-level (e.g. a Manager might have "Create PO" and "Receive Goods" but not "Approve PO").

11.5 Cross-Capability Data Relationships

Data relationships crossing capability boundaries degrade gracefully rather than error out (guest sales if Customer Management is off; hard dependency for PO→Supplier).

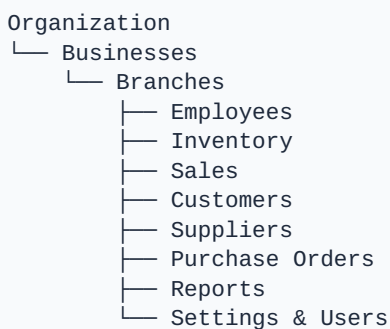
11.6 Extensibility Model

To add a new capability: define its six components, add it to the catalog and dependency table, assign it to relevant workspaces, mark it Available elsewhere. Navigation, permissions, dashboards, and notifications pick it up automatically.

11.7 Versioning a Capability

Each capability carries a version identifier so existing configurations migrate predictably when a capability's shape changes.

11A. Business Domain Model



This is the conceptual model underlying Business Structures (Section 7) and the Data Scope Model (Part VI) — those sections describe how this model is assigned (workspace at Business level) and secured (scoping at query level); this chapter is the entity model itself, independent of either concern.

11B. Business Rules

Business rules are architectural truths — invariants that hold regardless of implementation.

Products

- Every product belongs to one category.
- A product may have many variants.
- A barcode identifies one product.

Inventory

- Inventory exists per branch (per the Business Domain Model, 11A).
- Stock cannot be negative unless explicitly permitted.

- Every adjustment creates history (Inventory History, per IA Section 8).

Sales

- Every sale belongs to one branch.
- Completed sales are immutable (per Sales & Returns module spec, Part IV).
- Returns never delete sales — they create a linked Return record (see State Models, 11C).

Purchasing

- Every purchase order belongs to one supplier — a hard dependency (Sections 10.4, 11.5).
- Goods Receiving changes inventory; Approvals never change inventory (per the PO module spec's ripple-effects field, Part IV).

Employees

- Every employee belongs to at least one branch (per Permission Scoping, 12.3).
- Permissions are action-based, never capability-blanket (Section 13.1).
- Invitations expire automatically (see State Models, 11C).

Capabilities

- Disabling a capability never deletes data (Section 10.7).
- Locked capabilities preserve data (Section 10.5).

11C. State Models

Purchase Order

Draft → Submitted → Approved → Receiving → Partially Received → Completed / Cancelled

Partially Received is a first-class state, not transitional — a PO can sit here indefinitely if a supplier fulfills over multiple deliveries, matching the Manager journey (17.3) and the lifecycle-filter-tabs pattern (9.6).

Sale

Cart → Payment → Completed → Returned → Refunded

Void is a separate terminal state reachable from Completed, distinct from Returned/Refunded - Void reverses a sale outright (same-day, cashier error), while Returned/Refunded represents a customer-initiated reversal after the fact. Business Rules (11B) requires completed sales remain immutable and auditable, so these must not share one state.

Employee Invitation

Created → Sent → Pending → Accepted → Expired

Inventory Transfer

Created → In Transit → Received → Completed

11D. Event Model

This formalizes the "ripple effects" already noted per-module in Part IV into one traceable, consolidated view.

- **Sale Completed:** Sale Completed → Inventory Updated → Reports Updated → Customer Updated → Notification Created → Audit Log Written

- **Purchase Order Approved:** PO Approved → Manager Notified → Receiving Enabled
- **Product Archived:** Product Archived → Search Updated → Catalog Updated → Historical Sales Preserved

Every event chain here should trace back to a specific module spec's "cross-module ripple effects" field in Part IV — this chapter is the map; the module specs are the source of truth for each individual link.

11E. Information Ownership

Module	Owns
Products	Product catalog
Inventory	Stock quantities
Suppliers	Supplier records
Purchase Orders	Purchasing lifecycle
Customers	Customer information
Sales	Transactions
Reports	Owns nothing (reads data)
Dashboard	Owns nothing (summarizes data)

This confirms two things stated elsewhere: Reports as a "terminal consumer" with zero outward ripple effects (Part IV), and Dashboard as a pure lens (Part IV) — both now traceable to one authoritative ownership table.

12. User Roles

12.1 The Role Model

A role is a named collection of action-level permissions, scoped optionally to specific branches. Standard roles are simply pre-authored defaults; custom roles use the identical underlying structure.

12.2 Standard Roles

Role	Shell	Scope	Defining permission pattern
Owner	Owner Shell	Whole business, all branches	Full permissions across every enabled capability; cannot be limited by anyone else
Manager	Manager Shell	Assigned branch (es)	Full operational permissions within assigned branches; typically excluded from Approve PO, Business Configuration, Employee & Role Management unless granted
Cashier	Cashier Shell (POS)	One branch, often one till	Sales & Checkout only: Process Sale, Void Sale (often capped), Issue Refund (often capped)

12.3 Permission Scoping

Business vs. Branch

Every permission grant has two dimensions: what the role can do, and where it applies. Branch scope is stored per role-assignment, not per role definition. For multi-business Organizations, a person's access is to one Business first, then optionally scoped to branches within it — two separate role assignments for two businesses, never one spanning both.

12.4 Custom Roles

Authored from Employees → Roles: start from a template or blank slate, toggle action-level permissions capability by capability, name the role, optionally restrict to specific capabilities entirely.

12.5 Role Assignment & the Invitation Flow

Owners self-register with automatic Owner assignment. Every other role is invitation-only — an Owner or permitted Manager invites by email, assigning role and branch scope at invitation time. Invitation acceptance uses magic-link or temporary-password flow, never a plaintext password in the email.

12.6 Roles, Activity, and the Audit Trail

- **Activity** — What a specific employee did (operational, performance/troubleshooting).
- **Audit Log** — Business-wide, security-relevant events (role changes, permission edits, invitations, capability toggles, new-device logins).

13. Permission Model

13.1 Permission Structure

Capability → Permission → Action

Example - Inventory: Read, Create, Update, Delete, Transfer, Adjust, Approve.

13.2 Permission Conflicts & Precedence

- **Capability off always wins** — No role can grant access to a disabled capability.
- **Explicit branch scope always wins** over "all branches."
- A role can only grant permissions the assigning user themselves holds — prevents privilege escalation through role authoring.

13.3 Role Display in Navigation & UI

Where a person lacks permission for an action within an otherwise-visible screen, the UI shows the action visibly disabled with an explanation (e.g. "Owner approval required") if its absence would be confusing; routine actions entirely outside the role's job are omitted entirely rather than shown disabled.

14. Feature Availability Model

Workspace → Capabilities → Permissions → Navigation → Page → Component → Action

Nothing is hardcoded at any layer — a component (e.g. an "Approve" button) checks the resolved permission at render time, not a role name. This is what makes the disabled-vs-omitted distinction (13.3) possible.

PART III — Product Experience

15. UX Principles

Recognition · Feedback · Consistency · Error Prevention · Speed · Trust · Predictability.

16. Workspace Experiences

- **Owner Experience** — Goal: confidence and control across the whole business. Mental model: "how is my business doing, and what needs me." Primary tasks: review performance, approve, configure. Success metric: time-to-clear on Needs Attention.
- **Manager Experience** — Goal: keep today's operations running smoothly at their branch(es). Mental model: "is anything broken right now, and what's arriving." Primary tasks: monitor stock, receive deliveries, process returns. Success metric: issues resolved same-day.
- **Cashier Experience** — Goal: serve the customer in front of them, fast, without errors. Mental model: "this one transaction, right now." Primary tasks: sell, hold, resume, return. Success metric: transaction time, error rate.

16A. Product Roadmap (Expanded)

Maturity	Includes
MVP	Retail, POS, Inventory, Purchasing, Reports
Production Ready	Customer Loyalty, Analytics, Advanced Reports, Notifications
Business Suite	CRM, Accounting, Expenses, AI Insights
Enterprise	Manufacturing, Warehousing, Fleet, Advanced Security
Platform	Developer APIs, Marketplace, Third-party Apps, Automation

This supersedes the flatter MVP/Phase-2/Phase-3 sketch previously drafted for the Appendix's Future Roadmap — that entry now points here instead of duplicating a separate breakdown.

17. User Journey Maps

17.1 Format

Each journey: Persona & Goal · Trigger · Frequency · Steps (action → response → governing principle) · Friction Points & Mitigations · Success Definition.

17.2 Cashier Journey "A Single Sale, Start to Finish"

Persona & Goal: Cashier, complete a purchase as fast as possible with zero errors.

Trigger: Customer approaches the till.

Frequency: Dozens-hundreds per shift.

Step	Screen/Module	Action	System Response	Governing Principle
1	Ready to Sell	Scan/search items	Live Search surfaces matches instantly	Doherty Threshold
2	Shopping Cart	Items populate; quantity typed directly	Running total updates live; no VAT yet	Type-only quantity; VAT deferred to receipt
3	Customer Selection	Attach customer (optional)	Cart proceeds identically either way	Graceful degradation
4	Payment	Select M-Pesa/Cash/Card	VAT and final total now shown	Confidence Through Clarity
5	Receipt	Generated, printed/sent	Transaction complete	Predictable pattern
6	Ready to Sell	Cart clears	Ready for next customer	One primary next step

Friction Points: Connectivity drops (Offline Mode queues locally); wrong item added (removable, no confirm dialog Undo pattern); customer wants to pay later (Resume Held Sale); till lockout mid-shift (PIN model, doesn't lose held cart).

Success Definition: A trained cashier completes a typical sale near-instantly closer to muscle memory than "using software."

17.3 Manager Journey - "Morning Open + Receiving a Delivery"

Persona & Goal: Manager, open the store and process an incoming delivery against a PO.

Trigger: Start of shift; truck arrives.

Frequency: Daily (open); a few times weekly (deliveries).

Step	Screen/Module	Action	System Response	Governing Principle
1	Manager Dashboard	Review Needs Attention	Low-stock + PO-awaiting-receipt surfaced	Actionable vs. passive notifications
2	Inventory Status	Scan stock levels	Dense, scannable table	Module Anatomy consistency
3	Purchase Orders → Goods Receiving	Open relevant PO	Shows expected items/quantities	Recognition over recall
4	Goods Receiving	Mark received quantities	PO status updates automatically	Predictable Navigation
5	Inventory	Stock updates automatically	Low-stock alert clears if resolved	No manual double-entry
6	Manager Dashboard	Confirm Needs Attention clear	Dashboard reflects updated state	Confidence Through Clarity

Friction Points: Delivery doesn't match PO exactly (Partially Received is first-class, not an error); Manager lacks Approve PO (shown disabled with explanation); Manager oversees multiple branches (branch switch triggers unsaved-changes guard if mid-entry).

Success Definition: Manager starts the day knowing what needs attention, resolves it in the same module, never cross-references a paper PO from memory.

17.4 Owner Journey - "Weekly Business Review Across Branches"

Persona & Goal: Owner, understand weekly performance across branches and act on decisions.

Trigger: Owner's own routine, or pulled in by Needs Attention.

Frequency: Weekly (full); daily (lighter version).

Step	Screen/Module	Action	System Response	Governing Principle
1	Owner Dashboard	Review Business Performance, All Branches	Aggregate figures, default "this week"	Default to now
2	Needs Attention	See pending PO approval + subscription notice	Ranked by decision-importance	Von Restorff
3	Purchase Orders → Approvals	Approve PO	Moves to Approved; Manager's dashboard updates	Predictable Navigation
4	Reports → Profit Analysis	Filter to underperforming branch	Branch/filter state persists	Navigation Persistence

Step	Screen/Module	Action	System Response	Governing Principle
5	Employees → Activity	Check employee activity after anomaly	Shows that employee's actions only	Activity vs. Audit Log
6	Employees → Roles	Adjust Manager's permissions	Effective immediately, logged	Auditable role changes
7	Owner Dashboard	Return; Needs Attention clear	Reinforces control	Confidence Through Clarity

Friction Points: Multi-business Organization (context strip must make active business unmistakable); over-permissive grant (precedence rule caps risk, but UI should still confirm); unused capability revealed (nudge toward Workspace Architecture 10.7).

Success Definition: Owner leaves knowing exactly what happened, having acted on everything needing a decision, without leaving the Owner Shell.

17.5 Cross-Cutting Observations

- Notifications are the connective tissue between roles — same actionable-notification pattern, different scope.
- The same data event ripples across dashboards — module specs must document these explicitly.
- Every journey touches Employees/Roles — confirms Section 12 correctly precedes Journey Maps.

17A. Cross-Reference Matrix

Chapter	Depends On
Navigation	Workspace Architecture
Permissions	Capability Architecture
User Journeys	Navigation + Permissions
Modules	Capability Architecture
Design	Design System
Technical Foundation	SDD
Business Rules	Capability Architecture + Permission Model
State Models	Module Specifications
Event Model	Module Specifications (ripple-effects fields)

18. Screen Inventory (seed)

ID	Screen	Parent	Role(s)	Primary Action	Status
SCR-001	Owner Dashboard	Application	Owner	Review Needs Attention	Designed
SCR-002	Manager Dashboard	Application	Manager	Review Store Summary	Designed
SCR-003	POS → Ready to Sell	POS Workspace	Cashier	Add item to cart	Designed

ID	Screen	Parent	Role(s)	Primary Action	Status
SCR-004	POS → Payment	POS Workspace	Cashier	Complete Sale	Designed
SCR-005	Product List	Products	Owner, Manager	Add Product	Not started
SCR-006	Stock Adjustment	Inventory	Manager	Submit adjustment	Not started
SCR-007	Purchase Order → Create	Purchase Orders	Manager, Owner	Submit PO	Not started
SCR-008	Purchase Order → Approvals	Purchase Orders	Owner	Approve PO	Not started
SCR-009	Supplier Details	Suppliers	Owner, Manager	Edit Supplier	Not started
SCR-010	Customer Details	Customers	Owner, Manager, Cashier (view)	Edit Customer	Not started
SCR-011	Sale Details	Sales & Returns	Owner, Manager	Void / Return	Not started
SCR-012	Reports — Profit Analysis	Reports	Owner	Export Report	Not started
SCR-013	Employee List	Employees	Owner	Invite Employee	Designed
SCR-014	Business Settings — Backup & Data	Business Settings	Owner	Create/Restore Backup	Designed
SCR-015	Notification Center	Application	All	Resolve actionable item	Not started
SCR-016	User Profile — Security tab	User Profile	All	Update password/ 2FA	Designed
SCR-017	403 / 404 / 500 System	All	Return / retry	Designed	

This table becomes the literal project backlog as remaining screens are added and Status is tracked.

18A. Glossary (Expanded)

- **Organization** — The top-level account; may contain multiple Businesses.
- **Business** — A single operating entity with its own workspace and capability set.
- **Branch** — A physical or operational location under a Business, sharing its workspace.
- **Workspace** — A named, curated bundle of capabilities assigned per Business.
- **Capability** — A toggleable functional area representing a business activity.
- **Module** — The screen/IA unit implementing a capability.
- **Role** — A named collection of action-level permissions, standard or custom.
- **Permission** — A specific, grantable action within a capability.
- **Feature** — Always resolved to either a Capability (business-controlled) or a Feature Flag (platform-controlled) — never used as an undifferentiated third term.
- **Tenant** — An Organization, in multi-tenant architecture terms (Part VI).
- **Navigation Shell** — The role-specific navigation frame (Owner, Manager, Cashier).

- **Command Search** — The unified search-and-action surface (Section 9.10).
- **Dashboard** — The role-specific summarizing entry point; owns no data of its own.
- **Business Configuration** — The capability governing Business Settings.

19. Cross-Cutting Experiences

Search (Command Search) · Notifications (passive/actionable) · Help (contextual) · Offline (queue-and-sync) · Errors (403/404/500) · Loading (skeletons matching layout) · Empty States (teach the next action) · Success (toasts) · Printing/Exporting · Confirmation Dialogs (destructive-only; reversible actions get Undo) · Authentication · Session Expiry (preserve in-progress work, never silent logout).

19A. Product Governance

- **Architecture Decisions** — Major architectural decisions must be documented in the Architecture Decision Log (Appendix).
- **Design Changes** — New UI patterns must first update the Design System before appearing elsewhere — the rule already followed when the status-pill fix was made to the Design System before any screen was corrected, now stated explicitly as policy.
- **Capability Changes** — New capabilities must define: dependencies, permissions, navigation contribution, dashboard contribution, notification types — the six-component anatomy already defined in Section 11.1; this governance rule mandates it's followed every time.
- **Breaking Changes** — Must preserve existing business data and configurations wherever possible, consistent with the "disabling never deletes" rule (11B).

PART IV — Functional Modules

(Each follows: Purpose · Capabilities · Pages · Primary Workflows · Dependencies · Permissions · KPIs · Future Expansion)

Dashboard

Purpose: Owner/Manager entry point surfacing what happened and what needs a decision. **Capabilities:** Reads all enabled capabilities, owns none. **Pages:** Owner Dashboard, Manager Dashboard. **Workflows:** 17.3, 17.4. **Dependencies:** Every module. **Permissions:** View only; drill-through inherits target module. **KPIs:** Time-to-first-action, Needs Attention clear rate. **Future:** Customizable/reorderable cards.

Point of Sale

Purpose: Execute a sale, start to finish, as fast as possible. **Capabilities:** Sales & Checkout. **Pages:** Ready to Sell, Cart, Payment, Receipt, Held Sales, Offline States. **Workflows:** 17.2. **Dependencies:** Inventory, Customers, Products. **Permissions:** Process Sale, Void Sale, Issue Refund. **KPIs:** Average transaction time, error/void rate. **Future:** Restaurant-style order flow.

Products

Purpose: Maintain the catalog every other operational module points back to. **Capabilities:** Products & Inventory. **Pages:** Product List (standard anatomy), Product Details/Add/Edit (form anatomy). **Key entities:** Product, Category, Brand, Variant, Barcode. **Permissions:** Create/Edit/Archive Product, Manage Categories/Brands/Variants/Barcodes. **Notifications:** None directly. **Dependencies/ripple:** Archiving must not break historical Sales records; new Product auto-creates zero-stock Inventory record.

Inventory

Purpose: Track real-world stock per product per branch and its movements. **Capabilities:** Products & Inventory. **Pages:** Inventory Overview/Stock Levels (standard anatomy), Stock Adjustments/Stock Count (guided-flow anatomy). **Key entities:** Stock Level, Stock Adjustment, Stock Count, Stock Transfer. **Permissions:** Adjust Stock, Perform Stock Count, Transfer Stock (initiate/receive separately), View Valuation. **Notifications:** Actionable — Low Stock, Out of Stock; Passive — Count completed, Transfer received. **Ripple:** Goods Receipt increases stock; Sale decreases it; Transfer moves atomically between branches.

Suppliers

Purpose: Maintain who the business buys from, independent of orders placed. **Capabilities:** Supplier Management. **Pages:** Supplier List (standard), Supplier Details (form + tabs: Products, Purchase History, Performance). **Permissions:** Add/Edit Supplier, View Performance (view-only). **Ripple:** Deleting a supplier with open POs is blocked outright (hard dependency).

Purchase Orders

Purpose: Manage the ordering-to-receiving cycle. **Capabilities:** Purchasing. **Pages:** PO List (lifecycle-state filter tabs), Create PO (guided flow). **Permissions:** Create PO, Approve PO, Receive Goods, Cancel PO. **Notifications:** Actionable — Awaiting Approval, Awaiting Receipt; Passive — Approved, Received. **Ripple:** Approval doesn't touch Inventory (only Goods Receipt does); discrepant receipt flags Supplier Performance automatically.

Customers

Purpose: Maintain who the business sells to beyond a single transaction. **Capabilities:** Customer Management. **Pages:** Customer List (standard), Customer Details (form + tabs: Purchase History, Loyalty). **Permissions:** Add/Edit Customer, View Purchase History, Manage Loyalty. **Ripple:** Optional attachment from Sales & Returns per capability graceful-degradation rule.

Sales & Returns

Purpose: System of record for every completed transaction and its reversal. **Capabilities:** Sales & Checkout. **Pages:** Sales History (standard), Sale Details (read-mostly, Void/Return as explicit actions). **Permissions:** Process Sale, Void Sale, Issue Refund, Process Return. **Notifications:** Actionable — Void requiring approval; Passive — Sale completed, Return processed. **Ripple:** Decreases Inventory, feeds Reports and Customer Purchase History; Void/Return reversed but remains audit-visible.

Reports

Purpose: Turn raw activity into decision-ready views. **Capabilities:** Business Analytics.

Header → Filter Bar → Visualization → Data Table → Export/Schedule

(*Deliberate exception to standard anatomy*). **Permissions:** View Reports (scoped by branch/type), Export, Schedule. **Notifications:** Passive only — scheduled report ready. **Ripple:** None outward — Reports is a terminal consumer.

Employees

Purpose: Manage who has access and what they can do. **Capabilities:** Employee & Role Management. **Pages:** List, Details, Invite, Roles, Permissions, Activity, Access History. **Workflows:** 12.5, 12.4. **Dependencies:** Every capability. **Permissions:** Invite, Assign Role, Edit Permissions, View Activity. **KPIs:** Time-to-provision new hire. **Future:** Shift scheduling, time clock.

Business Settings

Purpose: Business-level configuration, the Tesler's-Law absorption point. **Capabilities:** Business Configuration. **Pages:** Profile, Workspace, Branches, Taxes & Finance, Receipts, Hardware, Integrations, Notifications, Security, Subscription, Audit Log, Backup & Data. **Dependencies:** Gates every other capability's availability. **Permissions:** Owner-only by default, delegable. **Future:** Multi-currency, multi-language.

Notifications

Purpose: Surface passive/actionable items across the system. Cross-cutting, not tied to one capability. **Pages:** Notification Center, Preferences. **Dependencies:** Every module that generates a notification. **Permissions:** View own; Owners may see business-wide. **KPIs:** Actionable-notification resolution time. **Future:** Push/SMS for offline-first branches.

Help

Purpose: Self-serve support without leaving context. **Pages:** Getting Started, Tutorials, Guides, Troubleshooting, Contact Support, System Status, Release Notes. **Dependencies:** None. **Permissions:** Universal. **KPIs:** Deflection rate. **Future:** In-app contextual tooltips.

User Profile

Purpose: Personal account management, distinct from business configuration. **Pages:** Personal Info, Security, Password, 2FA, Notification Preferences, Active Sessions, Appearance. **Dependencies:** Auth architecture. **Permissions:** Self-service, universal. **KPIs:** 2FA adoption rate. **Future:** Personal activity export.

PART V — Design

Rather than duplicating detail, this part links to the documents that already own it:

- **Information Hierarchy, Interaction Rules, Component Behaviour, Visual Design Principles** → BMS Design System v1.1 (color, typography, spacing, status pills, buttons).
- **Accessibility** → Design System Section 12, plus the contrast work done during the status-pill fix.

Design Review Checklist (new):

- Follows standard module anatomy or has a documented, justified exception
- One primary action is visually dominant
- All status indicators use the tinted-pill pattern
- Navigation Context strip present and accurate
- Degrades gracefully if any referenced capability is disabled
- Empty/loading/error states designed, not just the happy path
- Keyboard path exists for high-frequency actions
- Unsaved-changes guard present on any form/multi-step flow

PART VI — Technical Foundation

High-level only — this is where the Blueprint hands off to the SDD.

- **Configuration Model** — Capability enable/disable state lives at the Business level, resolved at request time.
- **Data Scope Model** — Every query implicitly scoped by Organization → Business → Branch → Role, mirroring the permission scoping mechanism exactly.

- **Multi-Tenant Architecture** — One platform instance serves many Organizations; tenant isolation at minimum at the Business level.
- **Scalability Strategy** — The capability/workspace model is itself the scalability strategy — new business types and capabilities extend the system without schema-breaking changes.

Appendix

- **Architecture Decision Log** — Captures decisions (e.g. the Success-color fix, Void-vs-Undo pattern choice) as they happen.
- **Naming Conventions** — e.g. always "Void," never "Cancel," for a completed sale reversal.
- **Future Roadmap / Release Phases** — See Section 16A (Product Roadmap) for the authoritative maturity-based breakdown.
- **Design Assets / Wireframe Index** — A living link table to Stitch outputs as they're built.

20A. Documentation Hierarchy (Final Chapter)

```

Business Management System (BMS)
├── Product Blueprint
│   ├── Product Vision
│   └── Architecture Decision Log
├── Software Requirements Specification (SRS)
├── Software Design Document (SDD)
│   ├── Domain Model
│   ├── Database Design
│   └── API Specification
├── Design Principles
│   └── Design System
├── Module Specifications
│   ├── POS Specification
│   ├── Products Specification
│   ├── Inventory Specification
│   ├── Purchasing Specification
│   ├── Reports Specification
│   ├── Employees Specification
│   └── Business Settings Specification
├── Wireframes
├── High-Fidelity Designs
└── Source Code

```

Note

The SDD already produced predates this Blueprint. Worth a pass at some point confirming nothing in the existing SDD contradicts decisions made here, particularly around the capability/permission model, which the SDD may have modeled differently before this architecture was fully worked out.

That's the complete Blueprint, start to finish. Ready to generate this as the Word document whenever you'd like — or if you want another review pass first, happy to hold off.